# Deprescribing Montelukast in Children with Asthma: A Systematic Review (SUPPLEMENTARY DATA SECTION)

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Tables: 3

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# Methods

*Table S1: Search Strategy*

| Strategy 918451 # | Database | Search term | Results |
|---|---|---|---|
| 1 | EMBASE | CHILD/ OR INFANT/ OR BABY/ OR NEWBORN/ | 2417196 |
| 2 | EMBASE | JUVENILE/ OR CHILD/ OR BOY/ OR GIRL/ OR INFANT/ OR TODDLER/ OR "SCHOOL CHILD"/ OR "PRESCHOOL CHILD"/ | 2301714 |
| 3 | EMBASE | JUVENILE/ OR ADOLESCENT/ | 1563364 |
| 4 | EMBASE | PEDIATRICS/ | 78300 |
| 5 | EMBASE | ((Paediatric* OR Infan* OR Newborn* OR Bab* OR Neonat* OR Preterm* OR Prematur* OR postmatur* OR Child* OR School* OR "Nursery school*" OR Kindergar* OR "Primary school*" OR "Secondary school*" OR "Elementary school*" OR "High school*" OR Highschool* OR Preschool* OR Kid* OR Toddler* OR Adoles* OR Teen* OR Boy* OR Girl* Minor* OR Pubert* OR Pubescen* OR Prepubescen*) NOT REVIEW*).ti,ab | 3217377 |
| 6 | EMBASE | "MINOR (PERSON)"/ | 653 |
| 7 | EMBASE | ADOLESCENCE/ OR PUBERTY/ OR PREPUBERTY/ | 70704 |
| 8 | EMBASE | (1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7) | 4839396 |
| 9 | EMBASE | 8 [Humans] | 4025357 |
| 10 | EMBASE | ASTHMA/ OR "ASTHMA BRONCHIALE"/ | 231772 |
| 11 | EMBASE | ((Asthma* OR Wheez* OR "Bronchial asthma*" OR "stable asthma") NOT REVIEW).ti,ab | 216313 |
| 12 | EMBASE | (10 OR 11) | 284696 |
| 13 | EMBASE | 12 [Humans] | 238534 |
| 14 | EMBASE | DEPRESCRIPTION/ OR DEPRESCRIPTIONS/ | 501 |
| 15 | EMBASE | "TREATMENT WITHDRAWAL"/ OR "DRUG WITHDRAWAL"/ | 215067 |
| 16 | EMBASE | ((deprescrib* OR stop* OR withdraw* OR ceas* OR discontinu* OR de-prescrib*) NOT REVIEW).ti,ab | 536525 |
| 17 | EMBASE | (14 OR 15 OR 16) | 680842 |
| 18 | EMBASE | 17 [Humans] | 519666 |
| 19 | EMBASE | MONTELUKAST/ | 9658 |
| 20 | EMBASE | ((montelukast OR singular) NOT REVIEW).ti,ab | 15377 |
| 21 | EMBASE | (19 OR 20) | 21572 |
| 22 | EMBASE | 21 [Humans] | 14849 |

| | | | |
|---|---|---|---|
| 23 | EMBASE | (8 AND 12 AND 17 AND 21) | 194 |
| 24 | EMBASE | 23 [Humans] | 191 |
| 25 | Medline | CHILD/ OR "CHILD, PRESCHOOL"/ | 1916435 |
| 26 | Medline | ADOLESCENT/ OR MINORS/ | 2036089 |
| 27 | Medline | INFANT/ OR "INFANT, NEWBORN"/ | 1139739 |
| 28 | Medline | ((Paediatric* OR Infan* OR Newborn* OR Bab* OR Neonat* OR Preterm* OR Prematur* OR postmatur* OR Child* OR School* OR "Nursery school*" OR Kindergar* OR "Primary school*" OR "Secondary school*" OR "Elementary school*" OR "High school*" OR Highschool* OR Preschool* OR Kid* OR Toddler* OR Adoles* OR Teen* OR Boy* OR Girl* Minor* OR Pubert* OR Pubescen* OR Prepubescen*) NOT REVIEW*).ti,ab | 2716250 |
| 29 | Medline | PUBERTY/ | 13325 |
| 30 | Medline | (25 OR 26 OR 27 OR 28 OR 29) | 4733651 |
| 31 | Medline | ASTHMA/ | 125895 |
| 32 | Medline | ((Asthma* OR Wheez* OR "Bronchial asthma*" OR "stable asthma") NOT REVIEW).ti,ab | 155279 |
| 33 | Medline | (31 OR 32) | 177833 |
| 34 | Medline | DEPRESCRIPTIONS/ | 466 |
| 35 | Medline | ((deprescrib* OR stop* OR withdraw* OR ceas* OR discontinu* OR de-prescrib*) NOT REVIEW).ti,ab | 369364 |
| 36 | Medline | (34 OR 35) | 369497 |
| 37 | Medline | ((montelukast OR singular) NOT REVIEW).ti,ab | 13162 |
| 38 | Medline | (30 AND 33 AND 36 AND 37) | 54 |
| 39 | Medline | 38 [Humans] | 50 |
| 40 | PubMed | ((Paediatric* OR Infan* OR Newborn* OR Bab* OR Neonat* OR Preterm* OR Prematur* OR postmatur* OR Child* OR School* OR "Nursery school*" OR Kindergar* OR "Primary school*" OR "Secondary school*" OR "Elementary school*" OR "High school*" OR Highschool* OR Preschool* OR Kid* OR Toddler* OR Adoles* OR Teen* OR Boy* OR Girl* Minor* OR Pubert* OR Pubescen* OR Prepubescen*) NOT REVIEW*).ti,ab | 143769 |
| 41 | PubMed | ((Asthma* OR Wheez* OR "Bronchial asthma*" OR "stable asthma") NOT REVIEW).ti,ab | 161822 |
| 42 | PubMed | ((deprescrib* OR stop* OR withdraw* OR ceas* OR discontinu* OR de-prescrib*) NOT REVIEW).ti,ab | 361553 |
| 43 | PubMed | ((montelukast OR singular) NOT REVIEW).ti,ab | 13080 |
| 44 | PubMed | (40 AND 41 AND 42 AND 43) | 0 |
| 45 | CINAHL | CHILD/ OR INFANT/ OR ADOLESCENCE/ OR "MINORS (LEGAL)"/ OR "CHILD, PRESCHOOL"/ | 892377 |
| 46 | CINAHL | PEDIATRICS/ OR NEONATOLOGY/ | 20791 |
| 47 | CINAHL | PUBERTY/ | 3060 |
| 48 | CINAHL | ADOLESCENCE/ OR CHILD/ OR "MINORS (LEGAL)"/ | 778755 |
| 49 | CINAHL | ((Paediatric* OR Infan* OR Newborn* OR Bab* OR Neonat* OR Preterm* OR Prematur* OR postmatur* OR Child* OR School* OR "Nursery school*" OR Kindergar* OR "Primary school*" OR "Secondary school*" OR "Elementary school*" OR "High school*" OR Highschool* OR Preschool* OR Kid* OR Toddler* OR Adoles* OR Teen* OR Boy* OR Girl* Minor* OR Pubert* OR Pubescen* OR Prepubescen*) NOT REVIEW*).ti,ab | 841196 |
| 50 | CINAHL | (45 OR 46 OR 47 OR 48 OR 49) | 1279403 |

| | | | |
|---|---|---|---|
| 51 | CINAHL | ASTHMA/ | 34317 |
| 52 | CINAHL | ((Asthma* OR Wheez* OR "Bronchial asthma*" OR "stable asthma") NOT REVIEW).ti,ab | 36601 |
| 53 | CINAHL | ((deprescrib* OR stop* OR withdraw* OR ceas* OR discontinu* OR de-prescrib*) NOT REVIEW).ti,ab | 73381 |
| 54 | CINAHL | (51 OR 52) | 45034 |
| 55 | CINAHL | MONTELUKAST/ | 368 |
| 56 | CINAHL | ((montelukast OR singular) NOT REVIEW).ti,ab | 1953 |
| 57 | CINAHL | (55 OR 56) | 2122 |
| 58 | CINAHL | (50 AND 53 AND 54 AND 57) | 14 |

## Results

*Table S2: Results of assessment of risk bias of randomised control trials| The Cochrane Collaboration's tool was used. Green= low risk of bias, Yellow= unclear risk of bias, Red= high risk of bias.*

| | **Kim *et al.*, 2020** | **Montuschi *et al.*, 2007** | **Kim *et al.*, 2005** |
|---|---|---|---|
| Random sequence generation (randomisation bias) | | | |
| Allocation concealment (randomisation bias) | | | |
| Baseline differences (randomisation bias) | | | |
| Participant awareness (deviation bias) | | | |
| Deliverer awareness (deviation bias) | | | |
| Appropriate analysis (deviation bias) | | | |
| Available data (data bias) | | | |
| Appropriate measure (measuring bias) | | | |
| Difference of measure (measuring bias) | | | |
| Assessor awareness (measuring bias) | | | |
| Pre-specified data analysis (reporting bias) | | | |
| Numerical analysis (reporting bias) | | | |
| Other bias | | | |

*Table S3: Results of assessment of risk bias of case-controlled and cohort studies| The Newcastle-Ottawa Quality Assessment Form was used. Green= star (low risk of bias), Yellow= unknown risk of bias, Red= No star (high risk of bias).*

| | **Lee *et al.*, 2005** | **Bratton *et al.*, 1999** |
|---|---|---|
| Representativeness of the exposed cohort (selection bias) | | |
| Selection of the non-exposed cohort (selection bias) | | |
| Ascertainment of exposure (selection bias) | | |
| Outcome of interest stated (selection bias) | | |
| Comparability bias | | |
| Assessment of outcome (outcome bias) | | |
| Appropriate follow-up of outcomes (outcome bias) | | |
| Appropriate follow-up of patients (outcome bias) | | |
| **Overall quality** | | |